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Co-jump dynamicity in the cryptocurrency market: A network modelling perspective

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ABSTRACT

We examine the co-jumps of 37 cryptocurrencies based on a network model, and analyse the portfolio implications. The results reveal that, firstly, Bitcoin exerts the strongest influence. Secondly, co-jump heterogeneity exists across pairs of cryptocurrencies with different market-capitalizations, and the impact of co-jumps is time-varying. Thirdly, the dynamic ranking of co-jump influence shows that, during the COVID-19 pandemic, Bitcoin dominates in the centrality ranking. However, smaller cryptocurrencies (Dogecoin and TRON) exhibit significant co-jump influence. Fourthly, the portfolios constructed based on the co-jump network outperform the baseline strategy by attaining higher returns while experiencing less volatility and shorter downside periods.

1. Introduction

Cryptocurrencies have received extensive attention from market participants in the past decade. The related literature focuses on the cryptocurrency markets in various aspects such as hedging properties (Bouri et al., 2017; Mensi et al., 2020), volatility modelling (Katsiampa et al., 2017; Ardia et al., 2019; Ghosh et al., 2023; Malek et al., 2023), risk spillovers (Ji et al., 2018; Yi et al., 2018; Akhtaruzzaman et al., 2022; Bouri et al., 2023), and price bubbling (Bouri et al., 2019; Ghosh et al., 2022). The cryptocurrency markets are highly speculative, relatively immature, and include many young and irrational participants (Almeida and Gonçalves, 2023). They exhibit a large dispersion of information and no consensus exits on how to put a value on a cryptocurrency (Bouri et al., 2019; Ma and Luan, 2022; Almeida and Gonçalves, 2023). These features make the market behaviour of cryptocurrencies dependant on sentiment and hype (Banerjee et al., 2022; Raimundo Júnior et al., 2022). Importantly, cryptocurrency prices experience violent fluctuations and periods of booms and busts, leading to enormous volatility, and the presence of jumps (Chaim and Laurini, 2018; Bouri et al., 2019; Shahzad et al., 2022; Zhang et al., 2022; Meng et al., 2023, 2023).

The literature on co-jumps amongst cryptocurrencies is very limited. Bouri et al. (2020) and Gkillas et al. (2022) find that cryptocurrencies exhibit co-jumping behaviour, which can trigger latter jumps.[1] Omrane et al. (2021) assess the significance of macroeconomic news for co-jumps in cryptocurrency markets. Except for Bouri et al. (2020) and Gkillas et al. (2022), there is no empirical evidence on how cryptocurrencies co-jump or the dynamic nature of their co-jumping behaviour. However, these two studies do not

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[1] Xu et al. (2022) detect jumps and co-jumps in blockchain and crypto-exposed US stocks and major cryptocurrencies. Phiromswad et al. (2021) show evidence of co-jumping activities between stock and cryptocurrency markets.

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consider the network of co-jumps nor the implications of co-jumps on portfolio management.[2] Specifically, the following questions remain unanswered: Which cryptocurrency can act as a bridge for jumps in the entire cryptocurrency market? Which cryptocurrency can hold the highest ranking in the co-jump network during normal and crisis periods? How do the dynamics of interrelated jumps in cryptocurrencies change during crisis events? What are the portfolio implications arising from evidence of co-jumps?

Our study is the first to offer answers to these important research questions. In this paper, we address these research gaps by identifying the jump transfer mechanism amongst 37 cryptocurrencies using a co-jump network[3] model and by examining the impact of co-jumps on portfolio allocation and performance. We apply the network model of Ding et al. (2023), which allows us to examine not only the occurrence of co-jumps in cryptocurrencies but also the intensity of co-jumps between pairs of cryptocurrencies from a network visualization perspective. Traditional co-jump methods focus on the co-jumps between two assets or cryptocurrencies, whereas the co-jump network model can simultaneously consider the relationships amongst multiple assets or cryptocurrencies. The co-jump network model captures more complex relationships amongst cryptocurrencies, which can reveal the common risk transmission paths and market structure amongst cryptocurrencies, aiding in a more accurate understanding of systemic risk and potential sources of shocks in the cryptocurrency markets.

We contribute to the literature by studying the (1) co-jump network, (2) dynamic ranking of co-jump influence over time, and (3) portfolio implications. Firstly, unlike the work of Bouri et al. (2020) and Gkillas et al. (2022), we built a co-jump network to learn the co-jump dependency of the cryptocurrency market, which can identify important nodes in the cryptocurrency network. By identifying these important nodes, market participants can gain a more accurate understanding of risk propagation paths and key drivers in the cryptocurrency market. Such contributions are essential for providing insight into cryptocurrency market interconnectivity, risk transfer, and dynamics.[4] Secondly, we estimate the dynamic ranking of co-jump influences, capturing the time-varying nature of co-jumps in the cryptocurrency market, extending the studies of Bouri et al. (2020) and Gkillas et al. (2022). Thirdly, we consider the portfolio implications of the presence of co-jumps by constructing better portfolios that diversify risk appropriately to cope with price jump events, which nicely complements the work of Bouri et al. (2020), Gkillas et al. (2022), Omrane et al. (2021), Phiromswad et al. (2021), and Papathanasiou et al. (2023), and contributes to improving portfolio optimization.

Our main findings reveal significant heterogeneity in co-jumps between cryptocurrencies with various market capitalizations. The impact of co-jumps varies over time, highlighting the dynamic nature of co-jump behaviour and transmission in the cryptocurrency market, which should provide valuable insights for traders and investors. Further analysis of the dynamic ranking of co-jump influence relationships amongst cryptocurrencies indicates that portfolio allocation can be better optimized, reducing risk and enhancing portfolio returns.

2. Data

Price data on 37 major cryptocurrencies at the daily and 5-minute frequencies are obtained from CoinMarketCap API[5] over the period May 15, 2018 - June 15, 2023.[6] Then daily log-returns are constructed as, $r_t = \log(p_t) - \log(p_{t-1})$, where p_t is the daily closing price of a cryptocurrency at time t, whereas 5-minute prices are used in subSection 3.1 to construct realized volatility(RV), realized bipower variation (BV), and the daily threshold level, which are then compared to the daily returns to identify whether a jump or a co-jump has occurred on that particular day.

Appendix Fig. A1 shows the daily prices of the 37 major cryptocurrencies over time, while Appendix Fig. A2 plots the log- returns (multiplied by 100). Appendix Table A1 presents the descriptive statistics of cryptocurrency daily returns. BNB and Dogecoin have the highest mean return, whereas Bytecoin has the lowest mean return. Bitcoin has the lowest standard deviation. Excess kurtosis is dominant, and skewness is negative in 25 cryptocurrencies. All return series are stationary.

3. Methods

To strike a balance between reducing the impact of microstructure noise and following Ding et al. (2023), we establish RV, BV, and daily threshold level using 5-minute price data to identify whether a jump or co-jump has occurred on that particular day.

[2] The concept of a co-jump network involves analysing and understanding the jump interdependencies between cryptocurrencies.

[3] Notably, the network model can reflect the dependent feature of edges in the real-world network, and discover pairwise jump dependency among a large number of cryptocurrencies.

[4] Specifically, we assess the intensity of co-jumps between cryptocurrencies based on the co-jump network, which helps us understand which cryptocurrencies have stronger impacts on others through jumps, and which have a weaker influence from other cryptocurrencies' jumps.

[5] Price data from coinmarketcap have been used by Bouri et al. (2019).

[6] There are two main reason for the selection of the sample period. Firstly, we want to capture the most recent data available on cryptocurrencies. Secondly, when constructing the co-jump network, it is important to include as many cryptocurrencies as possible. On the one hand, if the time period is too long, the sample size (i.e. number of cryptocurrencies) would be significantly reduced, failing to reflect the most important cryptocurrencies according to the recent data. This is because many of the cryptocurrencies used in our analysis were not available before 2018. On the other hand, if the time period is too short, it may compromise the reliability of our empirical analysis. Therefore, a five-year sample period is deemed reasonable, covering two major crisis events, the COVID-19 pandemic and the Russia-Ukraine conflict, which enriches our analysis.

3.1. Detecting jump and pairwise co-jump scores

Suppose we consider N logarithmic cryptocurrency price processes $\{Y_i(t)\}_{i=1}^{N}$, cryptocurrency *i* obeys the following process:

$$dY_i(t) = \mu_i(t)dt + \sigma_i(t)dW_i(t) + \kappa_i(t)dJ_i(t) \tag{1}$$

where $\mu_i(t)$ is the drift term of cryptocurrency *i*, $\sigma_i(t)$ is the volatility process of cryptocurrency *i*, $J_i(t)$ is the Poisson count process, and $\kappa_i(t)$ is the jump size process.

Assuming cryptocurrency prices at a frequency of *M*, we denote the *s*th intraday logarithmic return for cryptocurrency *i* on day *t* as:

$$r_{i,t,s} = Y_i\left(t - 1 + \frac{s}{M}\right) - Y_i\left(t - 1 + \frac{s-1}{M}\right), \; s = 1, 2, \ldots, M. \tag{2}$$

Under the physical measure, as *M* approaches infinity, the RV has the following approximate measure:

$$\lim_{M\to\infty} RV_{i,t} \to \int_{t-1}^{t} \sigma_i^2(u)du + \sum_{t-1<s\le t} \kappa_i^2(s), \tag{3}$$

where $\int_{t-1}^{t} \sigma_i^2(u)du$is the integrated volatility, and $\sum_{t-1<s\le t} \kappa_i^2(s)$ is the quadratic variation due to the jump component. Barndorff-Nielsen and Shephard (2004) propose the BV to estimate the integrated volatility consistently:

$$\lim_{M\to\infty} BV_{i,t} \to \int_{t-1}^{t} \sigma_i^2(u)du, \; BV_{i,t} = \frac{\pi}{2}\frac{M}{M-1}\sum_{s=1}^{M-1}\left|r_{i,t,s+1}\right|\left|r_{i,t,s}\right| \tag{4}$$

To identify price jumps, we use the non-parametric threshold method of Mancini (2009). Compared with other methods, such as Bouri et al. (2020) and Gkillas et al. (2022), non-parametric threshold estimation is less sensitive to model misspecification and outliers, making it more robust in the presence of noise and extreme observations, and allowing for the adaptive selection of threshold levels, which can vary over time and capture changes in the intensity and occurrence of jumps. Specifically, for each cryptocurrency *i*, we establish a daily threshold level:

$$\nu_{i,t} = \tau \times \left(\frac{1}{M}\right)^{0.49} \times \sqrt{\min\left(RV_{i,t}, BV_{i,t}\right)}, \tag{5}$$

where we set τ= 4 following Ding et al. (2023) and establish a daily threshold level. Assuming certain regularity conditions on $\mu_i(t)$ and $\sigma_i(t)$, and as the frequency *M* approaches infinity, consistent detection of jumps is possible (Mancini, 2009). The cumulative number of jumps detected for cryptocurrency *i* from day 1 to day *t* is:

$$J_{i,t} = \sum_{1\le d\le t} 1_{\left\{\left|r_{i,d}\right|>\nu_{i,d}\right\}}, \tag{6}$$

where $\{\mathbf{1}\}$ is the indicator function. Following Gilder et al. (2014), we define co-jumps between any pair of cryptocurrencies as the intersection of their individual jumps. We denote the cumulative number of co-jumps between cryptocurrency *i* and cryptocurrency *j* from day 1 to day *t* as:

$$CJ_{ij,t} = \sum_{1\le d\le t} 1_{\left\{\left|r_{i,d}\right|>\nu_{i,d}, \left|r_{j,d,s}\right|>\nu_{j,d}\right\}} \tag{7}$$

The co-jump indices between pairs of cryptocurrencies have three useful characteristics. Firstly, they are non-negative, where higher values indicate a stronger co-jump relationship. Secondly, they are scaled, allowing for comparisons between relationships amongst pairs of cryptocurrencies. Thirdly, they are symmetric, reflecting the symmetry of the co-jump relationship between the two cryptocurrencies.

3.2. Co-jump matrices and networks

Traditional co-jump methods focus on the co-jumps between two assets or cryptocurrencies, but cannot simultaneously consider the jump dependencies amongst multiple assets or cryptocurrencies. The co-jump network model can solve this problem, which helps reveal the co-jump relationships amongst cryptocurrencies and provide a more holistic understanding of the dynamic co-jumps of cryptocurrency markets.[7]

Using the pairwise co-jump measures, we build the corresponding daily $N \times N$ co-jump matrix A(T) based on Ding et al. (2023), as follows:

$$(A(T))_{ii} = \frac{J_i(T)}{T} \; for \; 1 \leq i \leq N, \; and \; (A(T))_{ij} = \frac{CJ_{ij}(T)}{T} \; for \; 1 \leq i \neq j \leq N. \tag{8}$$

Taking advantage of the symmetric co-jump matrices, we build co-jump networks to represent complex structures in the cryptocurrency markets. We consider dynamic co-jump networks of cryptocurrencies, $G_t = (V_t, E_t), t \in T$, where each vertex $N_{i,t} \in V$ represents a certain cryptocurrency at time t and a weighted edge $e_{i,j}(t) \in E$ denotes a type of co-jump relation between two cryptocurrencies at time t; the weight being the corresponding co-jump intensity. Co-jump matrices are used as representations of each co-jump network.

4. Empirical results

4.1. Results of jumps

Table 1 presents the statistics for the jumps observed from May 15, 2018 to June 15, 2023. Bitcoin, Ethereum, and BNB exhibit the lowest jump activity, with 244, 387, 387 jumps recorded, respectively. Conversely, Theta, Zilliqa, and Loopring account for the highest number of jumps, with 631, 631, and 656 jumps, respectively, representing 33.97%, 33.97%, and 35.32% of observed days. These findings demonstrate the recurring nature of jumps in these cryptocurrencies, particularly in mid-cap and low-cap cryptocurrencies. It is plausible to interpret these jumps as a consequence of lower liquidity and trading volume in these specific categories of cryptocurrencies (Bouri et al., 2020). Furthermore, our results indicate that cryptocurrencies holding dominant positions tend to exhibit comparatively fewer jumps, which bears significance for the modelling of volatility (Chen et al., 2019).

4.2. Co-jump network analysis

To provide a comprehensive visualization of co-jump relationships, we construct a network-based matrix, denoted as $A_{2018-2023}$, which captures all co-jump relations from May 15, 2018 to June 15, 2023. Fig. 1 shows the co-jump network based on the 37 cryptocurrencies. Each vertex in the graph represents an individual cryptocurrency, and the order of vertices corresponds to their market capitalization, arranged in a clockwise manner. Bitcoin, with the largest market capitalization, occupies the top position, while Quantum, with the smallest market capitalization, occupies the bottom position.

The co-jump network captures meaningful patterns in the cross-sectional structure of cryptocurrencies. Firstly, Bitcoin has the deepest red distribution, indicating its lower sensitivity to jumps from the other 36 cryptocurrencies. Secondly, cryptocurrencies with higher market capitalization generally have relatively lower sensitivity to jumps from other cryptocurrencies. The major colours of Ethereum, BNB, XRP, Dogecoin, TRON, and Monero are red or orange. Thirdly, cryptocurrencies within the medium and lower market capitalization ranges have a higher likelihood of experiencing co-jumps. The major colours of these cryptocurrencies are blue.

To analyse the impact of the co-jump network, we apply the approach of Mantegna (1999) to filter edges using a minimum spanning tree (MST) that maximizes the sum of the co-jump scores. Additionally, we use eigenvector centrality, based on the work of Bonacich (1987), to measure the influence of each cryptocurrency on the co-jump network. Fig. 2 shows the co-jump network after filtering the edges based on MST. We notice that Bitcoin has the strongest influence, indicating that the jumps in the other 36 cryptocurrencies are influenced by the jumps in Bitcoin. The colours of the edges connecting the other cryptocurrencies to Bitcoin, ranging from deep blue to deep red, reflect the varying degrees of influence that Bitcoin has on each cryptocurrency. Cryptocurrencies with edges closer to deep blue are more likely to be influenced by jumps in Bitcoin.

4.3. Dynamics ranking of co-jump influence

The rolling-sample ranking of the co-jump centrality analysis provides insight into the dynamics of the co-jump influence across individual cryptocurrency markets over time (from 2018 to 2023). Fig. 3 shows dynamics ranking of co-jump centrality by colours

[7] There are several aspects of the co-jump network, which can extend our understanding of co-jumps in cryptocurrency markets. Firstly, the co-jump network helps us identify which cryptocurrencies tend to co-jump together during price jumps. It allows us to understand the relationships and interdependencies between cryptocurrencies, which can provide valuable insights into market behavior. Secondly, the co-jump network analysis can shed light on the efficiency of the cryptocurrency markets. If co-jumps occur frequently and persistently, it may indicate inefficiencies or arbitrage opportunities that can be exploited by traders and investors. Finally, the co-jump network can be used to optimize cryptocurrency portfolios. By considering the jump interdependencies and correlations between cryptocurrencies, investors can construct portfolios that are better diversified and resilient to co-jump events.

Table 1
Statistics of jumps detected.

Rank by market capitalization	Cryptocurrency	Number of jumps	% of days with jumps	Rank by MC	Cryptocurrency	Number of jumps	% of days with jumps
1	**Bitcoin**	244	13.13%	20	**IOTA**	493	26.54%
2	**Ethereum**	387	20.84%	21	**Zcash**	523	28.16%
3	**BNB**	387	20.84%	22	**Dash**	451	24.28%
4	**XRP**	398	21.43%	23	**Loopring**	656	35.32%
5	**Cardano**	489	26.33%	24	**Zilliqa**	631	33.97%
6	**Dogecoin**	409	22.02%	25	**MCO**	478	25.74%
7	**TRON**	403	21.70%	26	**BAT**	562	30.26%
8	**Litecoin**	440	23.69%	27	**Nem**	429	23.10%
9	**Chainlink**	625	33.65%	28	**Qtum**	543	29.24%
10	**Monero**	404	21.75%	29	**Waves**	540	29.07%
11	**Ethereum Classic**	442	23.80%	30	**0x**	590	31.77%
12	**Stellar**	425	22.88%	31	**Ontology**	563	30.31%
13	**BCH**	433	23.31%	32	**Digibyte**	582	31.34%
14	**Vechain**	586	31.55%	33	**Steem**	493	26.54%
15	**EOS**	468	25.20%	34	**BitShares**	497	26.76%
16	**Tezos**	553	29.77%	35	**Lisk**	491	26.44%
17	**Theta**	631	33.97%	36	**Bytecoin**	598	32.20%
18	**Decentraland**	599	32.25%	37	**Quantum**	661	35.59%
19	**Neo**	500	26.92%				

Note: This table presents the number of jumps detected in the return series of the 37 cryptocurrencies under study, and the percentage (%) of days with jumps.

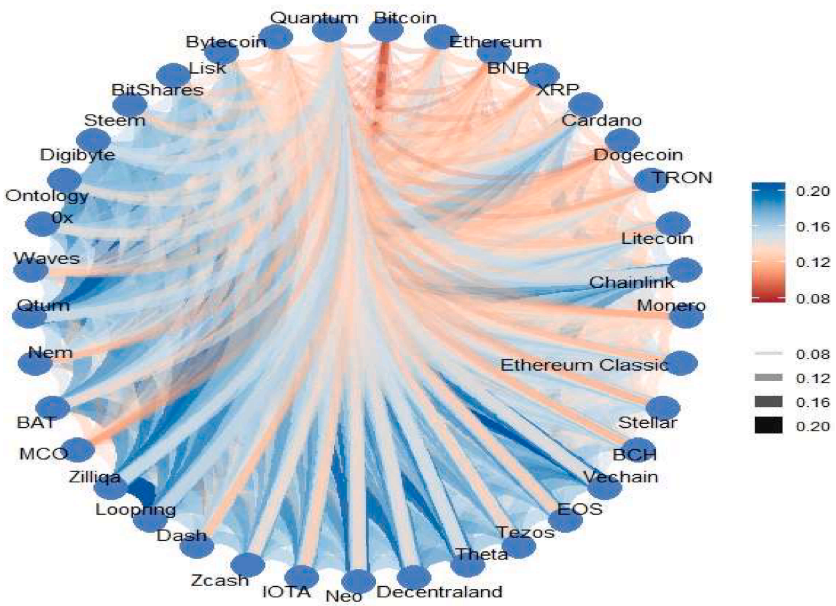

Fig. 1. Co-jump network for the period May 15, 2018 - June 15, 2023.
Notes: The edge connecting pairs of cryptocurrencies represent the intensity of the co-jump. In the colour bar (see the legend at top right), the colour distribution, ranging from deep blue to deep red, reflects the intensity of co-jump relationships between pairs of cryptocurrencies. In the colour bar (see the legend at bottom right), the colour distribution, ranging from light grey to black, reflects the width of the edge connecting the pairs of cryptocurrencies.

ranging from deep red to deep blue.

During the mid-2018 period, Bitcoin ranked highest in terms of net centrality. This suggests that Bitcoin's price surge during this period had a significant impact on other cryptocurrencies, making them more susceptible to jumps in Bitcoin. One potential explanation for this phenomenon is that the jumping behaviour of Bitcoin has amplification and incentive effects. During the COVID-19 pandemic, Bitcoin consistently maintained its dominant position in the centrality ranking, which can be attributed to several factors. Bitcoin, often compared to gold, is perceived as a safe haven during extreme events. The COVID-19 pandemic caused significant disruptions in global financial markets, and Bitcoin gradually gained public acceptance and accounted for the majority of the market

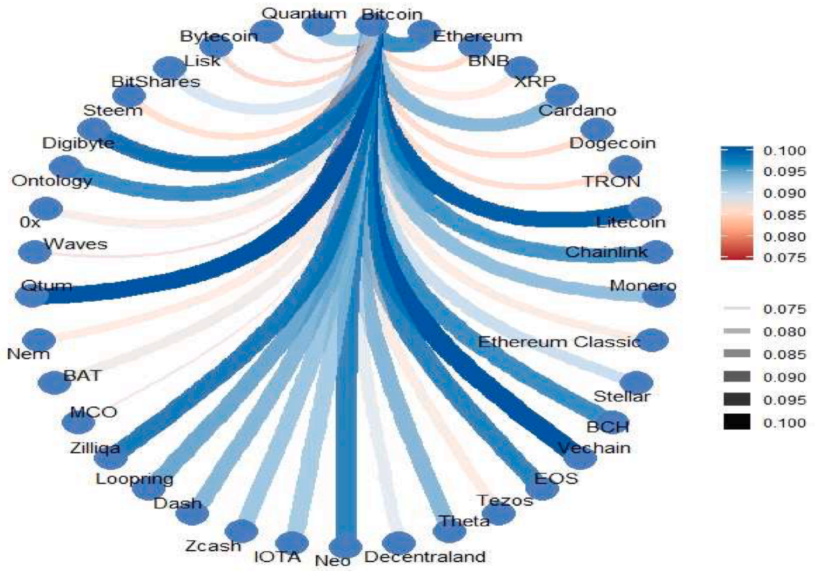

Fig. 2. Co-jump network based on MST for the full sample period.
Notes: See notes to Fig. 1.

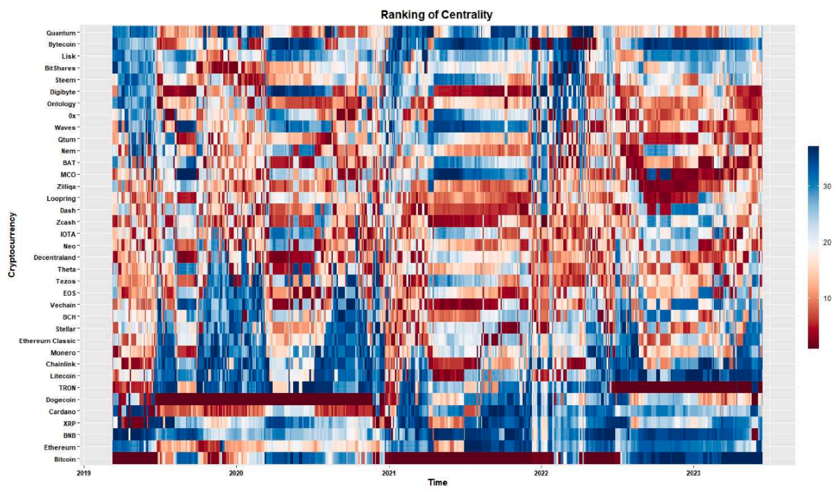

Fig. 3. Dynamic ranking of co-jump influence for the 37 cryptocurrencies.
Notes: In the colour bar (see the legend on the right), each colour stands for a ranking of co-jump influence, varying from 1 to 37. The deep red (blue) reflects high (low) ranking based co-jump network centrality.

capitalization in the crisis period. As a result, it became a representation of confidence in the cryptocurrency market, making it easier for jumps in other cryptocurrencies to be influenced by Bitcoin.

Unlike Bouri et al. (2020), we show that middle-sized cryptocurrencies are linked to other cryptocurrencies via the jump formation in larger cryptocurrencies such as Ethereum, BNB, and XRP. This influence remains stable over time, suggesting the importance of middle-sized cryptocurrencies in the formation of jumps in larger cryptocurrencies. These findings are generally in line with Bouri et al. (2019) and may be attributed to the reinforcement and contagion effects amongst major cryptocurrencies. By identifying the dynamics ranking of co-jump influence, cryptocurrency traders can leverage these effects and potentially profit by identifying jump dependencies between cryptocurrencies.

Overall, our results partially align with the findings of Bouri et al. (2020) and Gkillas et al. (2022) that jumps in one cryptocurrency impact the probability of jump occurrences in other cryptocurrencies. Different cryptocurrencies have different degrees of influence on other cryptocurrencies, which may be attributed to differences in market capitalization. Additionally, we observe that tail contagion in the cryptocurrency market is primarily observed in cryptocurrencies with medium and low market capitalization. This finding could be valuable for risk management and derivative pricing purposes for market participants, and also provide insight into the dynamics of co-jumps in the cryptocurrency market and their implications for portfolio management.

4.4. Application to portfolio allocation

Investors interested in constructing better portfolios that diversify risk appropriately to cope with price jump events (Arouri et al., 2019) can select cryptocurrencies with lower jump dependencies to achieve better portfolio performance. To demonstrate the economic value of the co-jump network in cryptocurrency markets, we develop two daily rebalanced mean-variance strategies based on the dynamics ranking of co-jump influence for each trading day over the period of study.

4.4.1. Mean-variance portfolio construction

We estimate mean-variance portfolio (MV) weights w_t by solving the following constrained optimization question:

$$\min_{w_t} w_t^T \widehat{\Sigma}_{t-1} w_t$$
$$\text{subject to } w_t^T \overrightarrow{1} = 1 \tag{9}$$
$$\| w_t \|_1 \leq l,$$

where $\widehat{\Sigma}_{t-1}$ denotes the covariance matrix, and $l \geq 0$ denotes the level of leverage.

The daily portfolio return can be calculated as:

$$r_t^{MV} = w_t^T r_t \tag{10}$$

where r_t is a vector of assect logarithmic returns.

The evaluation metric of portfolio performance is the Sharpe ratio. The annualized Sharpe ratio (Sharpe, 1998) is calculated as:

$$sr^{MV} = \frac{\overline{r^{MV}} \times 252}{\sigma^{MV}} \tag{11}$$

where $\overline{r^{MV}}$ denotes the average daily return of the mean-variance portfolio and σ^{MV} is the annualized volatility.

4.4.2. Analysis of portfolio performance

Here, we experiment with multiple mean-variance portfolio strategies and leverage constraints for optimization. Since we calculate mean-variance portfolio weights traded on day $t+1$ from covariance estimates from day $t = 360$ by rolling window, all of our analyses are conducted out-of sample.

The annualized Sharpe ratios of the mean-variance portfolios are presented in Table 2. According to the ranking of co-jump influence, portfolios built using the top five cryptocurrencies in terms of centrality have higher Sharpe ratios than the baseline mean-variance portfolio. Conversely, portfolios built using the bottom five cryptocurrencies have lower Sharpe ratios. As leverage constraints are relaxed, all mean-variance portfolios show an increasing annualized Sharpe ratio. We offer explanations for these findings. The selection of the top five cryptocurrencies, which have the most significant influence on the jumping behaviour of other cryptocurrencies while being less susceptible to such behaviour themselves, implies their relative independence from other cryptocurrencies. Consequently, when combined into a portfolio, these cryptocurrencies achieve improved risk diversification, thereby reducing portfolio volatility. In contrast, cryptocurrencies with lower centrality are more impacted by other cryptocurrencies, leading to increased portfolio volatility and uncertainty.

Additionally, we explore the trajectory of portfolio gains by plotting the cumulative returns of multiple mean-variance portfolio strategies in Fig. 4. The portfolios resulting from the co-jump network model outperform the baseline strategy by attaining higher returns while experiencing less volatility and shorter downside periods. Furthermore, we assess the robustness of co-jumps for portfolio management by considering various top cryptocurrencies and present the results in Appendix Table A2. The results show that considering the top cryptocurrencies outperforms the baseline strategy. However, the Sharpe ratio gradually decreases as more cryptocurrencies are added to the portfolio.

5. Conclusion

Frequent huge price spikes and declines in cryptocurrencies continue to attract the interest of traders and the investment community. Notably, the jumps and co-jumps in cryptocurrency prices have significant implications for risk management and portfolio allocation. This paper examines the co-jumps of cryptocurrencies and their transmission using a co-jump network model, which provides a new perspective for investigating the jump dependencies amongst a large number of cryptocurrencies.

Unlike Bouri et al. (2020) and Gkillas et al. (2022), our results show that Bitcoin jumps have the strongest influence on the jump formation in the other 36 cryptocurrencies. Co-jump heterogeneity exists amongst pairs of cryptocurrencies with different market capitalizations, and the impact of co-jumps varies over time. Our study complements the existing literature on the co-jump mechanisms of cryptocurrencies and provides new information for traders, investors, and miners. The research findings suggest that, when modelling the dynamic returns and portfolios of cryptocurrencies, it is necessary to consider not only whether co-jumps occur but also how they are transmitted and which cryptocurrencies have a strong co-jump impact on other cryptocurrencies. Furthermore, we

Table 2
Annualized Sharpe ratio of mean-variance portfolios.

Leverage	Baseline-MV	Top5 frequency rank-MV	Bot5 frequency rank-MV	Top5 mean rank-MV	Bot5 mean rank-MV
1	0.7948	**1.1099**	0.6307	0.8128	0.7651
2	0.8162	**1.1275**	0.6453	0.8339	0.7831
3	0.8234	**1.1334**	0.6502	0.8409	0.7891
4	0.8269	**1.1363**	0.6526	0.8444	0.7921
5	0.8291	**1.1380**	0.6540	0.8465	0.7939
6	0.8305	**1.1392**	0.6551	0.8480	0.7951
7	0.8315	**1.1401**	0.6557	0.8490	0.7960
8	0.8323	**1.1407**	0.6562	0.8497	0.7966
9	0.8329	**1.1412**	0.6566	0.8503	0.7971
10	0.8334	**1.1415**	0.6570	0.8508	0.7975

Notes: This table shows the annualized Sharpe ratio of mean-variance portfolios under different constrained. leverage.[81]

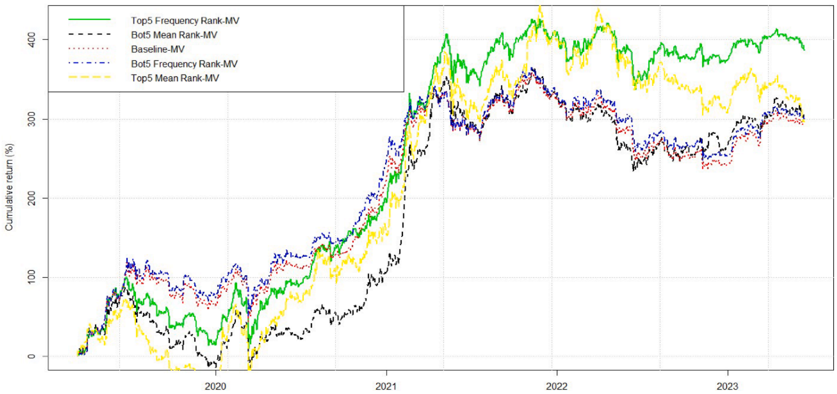

Fig. 4. Cumulative returns of portfolios.

provide practical implications by studying the mean-variance portfolio strategies based on the dynamics ranking of co-jump influence, which are found to achieve higher Sharpe ratios in comparison to the baseline portfolio strategy. Finally, our study has significance for both short-term and long-term investors.[9]

Our study has some limitations. Firstly, it overlooks the positive and negative components of co-jumps (see, Da Fonseca and Ignatieva, 2019) and their propagation across cryptocurrencies, which should be addressed in future studies. Secondly, we assume that the transaction costs for all portfolios are constant, which is too ideal, but provides some reference value for the practical implications of our portfolio analysis. Further research should evaluate portfolio performance under varying transaction costs and develop rebalancing strategies in a more realistic way. Future studies should also explore the relationship between co-jumps and spillover effects (El Khoury et al., 2023), and consider co-jumps between financial and nonfinancial firms (Akhtaruzzaman et al., 2021).

CRediT authorship contribution statement

Lei Zhang: Data curation, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Elie Bouri:** Visualization, Validation, Writing – original draft, Writing – review & editing. **Yan Chen:** Conceptualization, Data curation, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Validation.

Data availability

The authors do not have permission to share data.

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[9] Short-term investors may benefit from incorporating co-jump centrality rankings into their trading strategies to capture potential opportunities. Long-term investors can focus on cryptocurrencies that exhibit lower co-jump dependence, potentially reducing their exposure to systemic risks and enhancing their portfolio stability over time.

Appendix

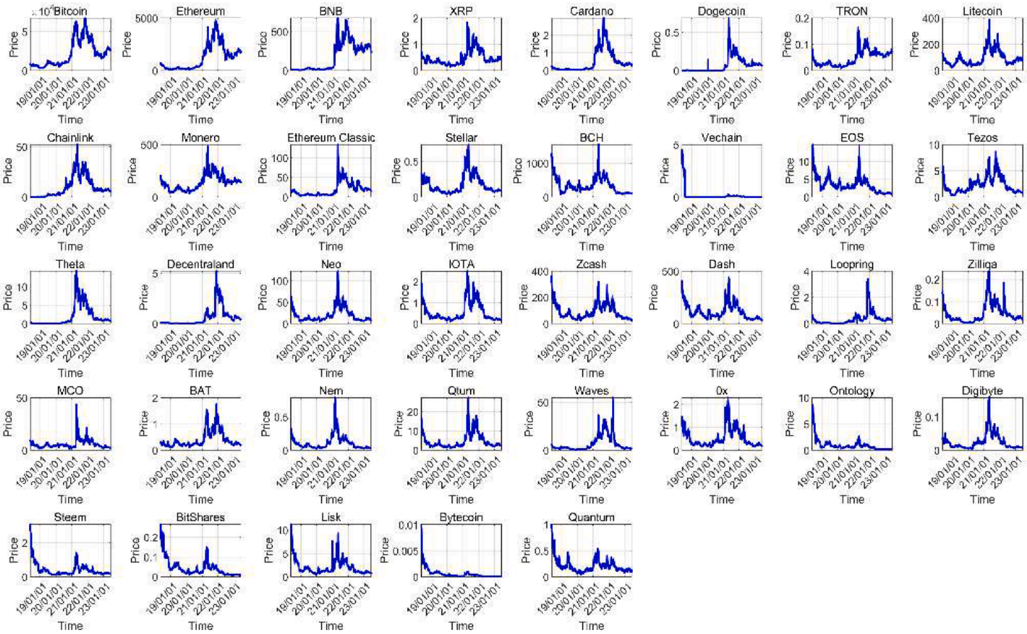

Fig. A1. Plots of price levels of cryptocurrencies.

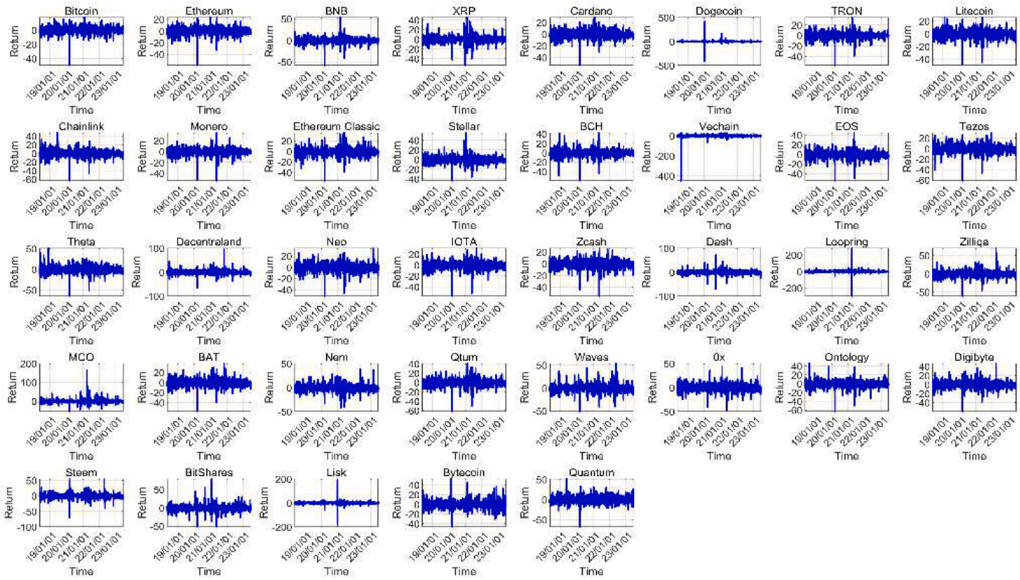

Fig. A2. Plots of return levels of cryptocurrencies.

[8] Baseline-MV indicates a portfolio strategy that considers all cryptocurrencies. Top5 Frequency Rank-MV indicates a portfolio strategy that considers the top five cryptocurrencies in terms of frequency ranking. Bot5 Frequency Rank-MV indicates a portfolio strategy considering the bottom five cryptocurrencies in terms of frequency ranking. Top5 Mean Rank-MV denotes a portfolio strategy considering the top five cryptocurrencies in terms of mean ranking. Bot5 Mean Rank-MV denotes a portfolio strategy considering the bottom five cryptocurrencies in terms of mean ranking.

Table A1
Statistical properties of daily returns.

	Mean	Max	Min	SD	Skewness	Kurtosis	P-value ADF test	Market Cap
Bitcoin	0.0584	17.7424	-49.7278	3.7235	-1.3229	22.3046	0.000	$484.54B
Ethereum	0.0457	23.0772	-58.9639	4.9162	-1.1987	17.0049	0.000	$197.10B
BNB	0.1585	53.0574	-58.1158	5.1799	-0.3913	22.2265	0.000	$36.31B
XRP	-0.0192	44.8991	-54.1017	5.6193	0.0909	17.9989	0.000	$24.70B
Cardano	0.0011	28.6973	-53.7199	5.5345	-0.334	9.9666	0.000	$8.99B
Dogecoin	0.1391	414.1786	-415.589	15.5341	0.5513	555.1136	0.000	$8.51B
TRON	0.0031	34.0529	-57.085	5.2278	-0.7991	15.0271	0.000	$6.30B
Litecoin	-0.0347	25.8175	-48.6778	5.1716	-0.8169	12.4284	0.000	$5.38B
Chainlink	0.1304	47.5424	-63.7153	6.6962	-0.3203	11.3856	0.000	$2.70B
Monero	-0.0232	34.2203	-53.5391	4.9961	-1.3242	19.3433	0.000	$2.43B
Ethereum Classic	-0.0109	35.5123	-56.1411	5.9082	-0.1398	13.7569	0.000	$2.09B
Stellar	-0.0787	55.3585	-44.0312	5.3667	0.482	15.9455	0.000	$2.07B
BCH	-0.1378	42.4173	-59.7721	5.8444	-0.5152	18.519	0.000	$2.00B
Vechain	-0.3073	29.3544	-447.938	12.2295	-26.4986	968.5455	0.000	$1.11B
EOS	-0.162	44.5821	-54.4567	5.8859	-0.6819	14.134	0.000	$694.14M
Tezos	-0.1014	30.1886	-61.4465	6.2397	-0.7791	12.3195	0.000	$665.77 M
Theta	0.053	50.9332	-64.773	6.8552	-0.2879	12.1889	0.000	$626.90 M
Decentraland	0.0504	95.042	-67.0547	7.1806	1.293	27.294	0.000	$614.98M
Neo	-0.1126	33.8645	-51.1497	5.8466	-0.6609	10.8654	0.000	$538.83 M
IOTA	-0.1378	30.8523	-57.1459	5.7349	-0.8523	13.8537	0.000	$413.66 M
Zcash	-0.1435	26.5126	-55.1113	5.7559	-0.8127	11.1784	0.000	$398.74M
Dash	-0.1443	71.1032	-70.57	6.1604	-0.0879	29.792	0.000	$330.61 M
Loopring	-0.0648	281.0026	-298.824	12.0288	-1.2782	367.4978	0.000	$277.88M
Zilliqa	-0.1189	69.5703	-61.0607	6.9271	0.4012	16.5597	0.000	$263.69M
MCO	-0.0977	164.0493	-54.2551	8.3948	4.9589	92.6693	0.000	$252.53 M
BAT	-0.0403	37.7259	-57.69	6.0258	-0.2636	11.3497	0.000	$243.54M
Nem	-0.1383	38.9918	-42.0962	5.7588	-0.1629	11.0391	0.000	$229.49 M
Qtum	-0.1138	40.5166	-62.0511	6.3019	-0.5645	13.837	0.000	$213.08M
Waves	-0.0829	53.5332	-51.0476	6.6964	0.4341	12.8416	0.000	$147.28 M
0x	-0.1175	43.324	-48.1002	6.3009	0.0233	9.7396	0.000	$142.21M
Ontology	-0.2087	45.5721	-63.2706	6.3513	-0.6949	14.5634	0.000	$140.57M
Digibyte	-0.0987	46.2886	-59.6144	6.5947	0.042	11.1408	0.000	$101.25 M
Steem	-0.1591	54.6815	-69.7052	6.4256	-0.4015	19.6585	0.000	$66.66M
BitShares	-0.187	78.948	-51.9311	6.6049	0.6757	24.4296	0.000	$22.92M
Lisk	-0.1476	189.6333	-184.804	8.4565	0.2189	262.2782	0.000	$15.25M
Bytecoin	-0.3002	52.3995	-46.4306	6.7739	0.4658	9.9867	0.000	$6.54M
Quantum	-0.1271	51.4777	-67.9333	7.1102	0.0935	10.6191	0.000	$6.32 M

Notes: This table reports summary statistics of daily returns of the 37 major cryptocurrencies over the sample period May 15, 2018 - June 15, 2023. It includes, the mean of returns, standard deviation (SD), largest and smallest return observations (denoted by max and min, respectively), skewness, and kurtosis. It also shows the p-value of the Augmented Dickey-Fuller (ADF) test. The last column shows the USD market capitalization of each cryptocurrency.

Table A2
Robustness test for annualized Sharpe ratio of mean-variance portfolios.

Leverage	Baseline-MV	Top5 Frequency Rank-MV	Top10 Frequency Rank-MV	Top15 Frequency Rank-MV	Top20 Frequency Rank-MV
1	0.7948	**1.1099**	0.9556	0.8296	0.8177
2	0.8162	**1.1275**	0.9741	0.8488	0.8369
3	0.8234	**1.1334**	0.9802	0.8552	0.8433
4	0.8269	**1.1363**	0.9833	0.8584	0.8465
5	0.8291	**1.1380**	0.9851	0.8603	0.8485
6	0.8305	**1.1392**	0.9863	0.8616	0.8497
7	0.8315	**1.1401**	0.9872	0.8625	0.8507
8	0.8323	**1.1407**	0.9879	0.8632	0.8513
9	0.8329	**1.1412**	0.9884	0.8637	0.8519
10	0.8334	**1.1415**	0.9888	0.8642	0.8523

Note: This table shows the robustness test for annualized Sharpe ratio of mean-variance portfolios under various constraints, which consider the top cryptocurrencies in the portfolio strategy.

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